1 Integration Formulas

To facilitate writing formulas for specific functions f, we now introduce the notation

$$\int_a^b f(x)dx$$

to mean the same thing as $\int_a^b f$. At the moment, the 'dx' doesn't have any independent meaning outside of the expression $\int_a^b f(x)dx$. (However, we will soon see that it can be interpreted as a differential, as the notation suggests.)

The Fundamental Theorem of Calculus says that if f is continuous on $[a,b]$, then its definite integral is given by $\int_a^b f(x)dx = F(b) - F(a)$, where F is any antiderivative of f. By a corollary of the Mean Value Theorem that we proved previously, any two antiderivatives for a function f on $[a,b]$ can only differ by a constant. We use the notation $\int f(x)dx$ (without limits on the integral sign), called an *indefinite integral*, to stand for an arbitrary antiderivative of f. If $F(x)$ is an antiderivative for $f(x)$, then we write $\int f(x)dx = F(x)+C$, where C is an arbitrary real constant. [1]

All of the derivative formulas we have learned so far yield antiderivative formulas simply by viewing them in reverse. These are recorded in the table on the final page. When using the formulas in this table to evaluate a definite integral $\int_a^b f(x)dx = F(b) - F(a)$, the choice of C does not matter and therefore C may be set to zero.

The following properties are useful for evaluating indefinite integrals of more complicated functions.

Theorem 1.1 (Properties of indefinite integrals)**.** *Let F and G be antiderivatives of f and g, respectively, and let k be any real constant. Then*

(a) $\int (f(x)+g(x))dx = F(x)+G(x)+C$

(b) $\int kf(x)dx = kF(x)+C$

Corollary 1.2. *If f and g are continuous functions on $[a,b]$ and k a real constant, then*

(a) $\int_a^b (f(x)+g(x))dx = \int_a^b f(x)dx + \int_a^b f(x)dx.$

(b) $\int_a^b kf(x)dx = k\int_a^b f(x)dx.$

Corollary 1.2 is actually true for any integrable functions f and g on $[a,b]$ (even if these are not continuous), which we will prove later.

[1] It would make more sense to define $\int f(x)dx$ to be the *set* of *all* antiderivatives of f; i.e., $\int f(x)dx = \{F(x)+C : C \in \mathbb{R}\}$, but this is not what calculus books do. Also, know that you will surely lose points on the AP exam if you forget to write the '$+C$' after finding an indefinite integral. Similarly, you will almost certainly lose points if you forget to write the 'dx' in $\int f(x)dx$ or $\int_a^b f(x)dx$.

1. Prove Theorem 1.1.

2. Prove Corollary 1.2.

3. Show that it follows from Theorem 1.1 that $\int (f(x)-g(x))dx = F(x)-G(x)+C$ and that, if f and g are continuous on $[a,b]$, then $\int_a^b (f(x)-g(x))dx = \int_a^b f(x)dx - \int_a^b g(x)dx$.

Using Theorem 1.1, we can evaluate any integral involving a sum, difference, or constant multiple of functions with known antiderivatives.

4. Evaluate each of the following indefinite integrals:

   (a) $\int (5x^3 - 7x^2 + 3x + 4)dx$

   (b) $\int \frac{x^2 + 4\sqrt[3]{x}}{x} dx$

   (c) $\int \frac{4}{1+x^2} dx$

   (d) $\int \tan(x)\cos(x)dx$

Name:_____

Differentiation Formula	Indefinite Integral
$\frac{d}{dx}(k) = 0$	$\int k\,dx = \int kx^0\,dx = kx + C$
$\frac{d}{dx}(x^n) = nx^{n-1}$	$\int x^n\,dx = \frac{x^{n+1}}{n+1} + C$ for $n \neq -1$
$\frac{d}{dx}(\ln\lvert x\rvert) = \frac{1}{x}$	$\int \frac{1}{x}\,dx = \ln\lvert x\rvert + C$
$\frac{d}{dx}(e^x) = e^x$	$\int e^x\,dx = e^x + C$
$\frac{d}{dx}(\sin x) = \cos x$	$\int \cos x\,dx = \sin x + C$
$\frac{d}{dx}(\cos x) = -\sin x$	$\int \sin x\,dx = -\cos x + C$
$\frac{d}{dx}(\tan x) = \sec^2 x$	$\int \sec^2 x\,dx = \tan x + C$
$\frac{d}{dx}(\csc x) = -\csc x \cot x$	$\int \csc x \cot x\,dx = -\csc x + C$
$\frac{d}{dx}(\sec x) = \sec x \tan x$	$\int \sec x \tan x\,dx = \sec x + C$
$\frac{d}{dx}(\cot x) = -\csc^2 x$	$\int \csc^2 x\,dx = -\cot x + C$
$\frac{d}{dx}\left(\sin^{-1} x\right) = \frac{1}{\sqrt{1-x^2}}$	$\int \frac{1}{\sqrt{1-x^2}} = \sin^{-1} x + C$
$\frac{d}{dx}\left(\tan^{-1} x\right) = \frac{1}{1+x^2}$	$\int \frac{1}{1+x^2}\,dx = \tan^{-1} x + C$
$\frac{d}{dx}\left(\sec^{-1}\lvert x\rvert\right) = \frac{1}{x\sqrt{x^2-1}}$	$\int \frac{1}{x\sqrt{x^2-1}}\,dx = \sec^{-1}\lvert x\rvert + C$